

Mohammad Ratul Mahjabin

mohammadratul@usf.edu | (813) 3605597 | www.ratul-mahjabin.github.io

Research Interests

Computational Social Science, Natural Language Processing, Social Media Mining, Machine Learning, Data Science, Human-Computer Interaction, Network Science, Computer Vision

Education

University of South Florida, Tampa, FL

Ph.D. in Computer Science and Engineering

Jan '24 - Present

Research Areas: Computational Social Science, Human-Computer Interaction, NLP

Advisor: Dr. Raiyan Abdul Baten - [Computational Social Science and AI Lab](#)

Islamic University of Technology, Dhaka, Bangladesh

B.Sc. in Software Engineering

Jan '19 - May '23

Thesis: [An Efficient Deep Learning-based approach for Recognizing Agricultural Pests in the Wild](#)

Publications

Journal Papers

- J3 Mustafina SN, Islam MN, **Mahjabin MR**, Mannan MM, Islam MM. [Identifying key indicators to develop a novel mobile application for early screening of postpartum depression in developing countries](#). BMC Health Services Research. 2025 Dec;25(1):1-20.
- J2 Rafi TH, **Mahjabin R**, Ghosh E, Ko YW, Lee JG. [Domain generalization for semantic segmentation: a survey](#). Artificial Intelligence Review. 2024 Aug 12;57(9):247.
- J1 Islam MN, Jahangir R, Mohim NS, Wasif-UI-Islam M, Ashraf A, Khan NI, **Mahjabin MR**, Miah AS, Shin J. [A multilingual handwriting learning system for visually impaired people](#). IEEE Access. 2024 Jan 15.

Conference Papers

- C3 Baten, R., & **Mahjabin, M.** (2025). [The Wisdom of Intellectually Humble Networks](#). Proceedings of the Annual Meeting of the Cognitive Science Society, 47. Retrieved from <https://escholarship.org/uc/item/1hd2t78f>
- C2 Hema NE, Zeest NS, **Mahjabin MR**, Islam MN. [An Ensemble-based ML Model to Predict Cardiac Disease](#). In 2024 International Conference on Inventive Computation Technologies (ICICT) 2024 Apr 24 (pp. 214-219). IEEE
- C1 Rafi MH, **Mahjabin MR**, Rahman MS, Kabir MH, Ahmed S. [A Critical Analysis of Deep Learning Applications in Crop Pest Classification: Promising Pathways and Limitations](#). In 2023 26th International Conference on Computer and Information Technology (ICCIT) 2023 Dec 13 (pp. 1-6). IEEE.

Professional Experience

University of South Florida , Tampa, FL <i>Graduate Research and Teaching Assistant, Dept. of Computer Science and Engineering</i>	<i>Jan '24 - Present</i>
Military Institute of Science and Technology (MIST) , Dhaka, Bangladesh <i>Research Assistant, Dept. of Computer Science and Engineering</i>	<i>Aug '23 - Dec '23</i>
Samsung R&D Institute , Dhaka, Bangladesh Software Engineer Intern	 <i>May '22 - Sept '22</i>

Selected Projects

PPD Coach An android application for Screening Postpartum Depression in the context of Bangladesh - It screens postpartum depression using story-based questions and visual aids. - Developed with: <u>Flutter</u>, <u>Firebase</u>	<i>Mobile Application</i> <i>2023</i>
ScrapeText Extracting articles from various online sources using Python <u>[Github]</u> - Text scraping tool designed to extract and perform textual analysis calculating different metrics - Developed with: <u>Python</u>, <u>Pandas</u>, <u>NLTK</u>, <u>Regex</u>	<i>Notebook</i> <i>2023</i>
FoodScan101 Enables users to detect and identify different types of food from a photo <u>[Github]</u> - A CNN(EfficientNetV2) was fine-tuned through transfer learning on a <u>dataset</u> - Developed with: <u>Python</u>, <u>Flask</u>, <u>CNN</u>	<i>WebApp</i> <i>2023</i>
HaateKhor ML-powered web app for early childhood education <u>[Github]</u> - This project utilizes ML techniques to provide early education to children in an engaging and interactive way <u>[Video Demo]</u> - Developed with: <u>Python</u>, <u>OpenCV</u>, <u>YOLO</u>, <u>React</u>, <u>Nodes.js</u>	<i>WebApp</i> <i>2021</i>
Covid-CXRNet Detecting Covid19 with Chest X-ray using CNN <u>[Github]</u> - This project project implements a CNN model(ResNet18) to accurately detect COVID-19 with an accuracy of 99.98% - Developed with: <u>Python</u>, <u>Pytorch</u>	<i>Notebook</i> <i>2021</i>

Skills

Programming	Python, R, C/C++, Javascript, Java, HTML, CSS
Libraries	PyTorch, TensorFlow, HuggingFace, NetworkX, NLTK, Pandas, Scikit-Learn, Matplotlib, Numpy, Spacy, Spicy
Web & DB	Flask, Django, React, MongoDB, Firebase, REST API
Tool	Docker, Heroku, Git, LaTeX